

Role of mitochondria in regulating immune response during bacterial infection.

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Mitochondria are dynamic organelles of eukaryotes involved in energy production and fatty acid oxidation. Besides maintaining ATP production, calcium signaling, cellular apoptosis, and fatty acid synthesis, mitochondria are also known as the central hub of the immune system as it regulates the innate immune pathway during infection. Mitochondria mediated immune functions mainly involve regulation of reactive oxygen species production, inflammasome activation, cytokine secretion, and apoptosis of infected cells. Recent findings indicate that cellular mitochondria undergo constant biogenesis, fission, fusion and degradation, and these dynamics regulate cellular immuno-metabolism. Several intracellular pathogens target and modulate these normal functions of mitochondria to facilitate their own survival and growth. De-regulation of mitochondrial functions and dynamics favors bacterial infection and pathogens are able to protect themselves from mitochondria mediated immune responses. Here, we will discuss how mitochondria mediated anti-bacterial immune pathways help the host to evade pathogenic insult. In addition, examples of bacterial pathogens modulating mitochondrial metabolism and dynamics will also be elaborated. Study of these interactions between the mitochondria and bacterial pathogens during infection will lead to a better understanding of the mitochondrial metabolism pathways and dynamics important for the establishment of bacterial diseases. In conclusion, detailed studies on how mitochondria regulate the immune response during bacterial infection can open up new avenues to develop mitochondria centric anti-bacterial therapeutics.

An Overview: The Diversified Role of Mitochondria in Cancer Metabolism.

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Mitochondria are intracellular organelles involved in energy production, cell metabolism and cell signaling. They are essential not only in the process of ATP synthesis, lipid metabolism and nucleic acid metabolism, but also in tumor development and metastasis. Mutations in mtDNA are commonly found in cancer cells to promote the rewiring of bioenergetics and biosynthesis, various metabolites especially oncometabolites in mitochondria regulate tumor metabolism and progression. And mutation of enzymes in the TCA cycle leads to the unusual accumulation of certain metabolites and oncometabolites. Mitochondria have been demonstrated as the target for cancer treatment. Cancer cells rely on two main energy resources: oxidative phosphorylation (OXPHOS) and glycolysis. By manipulating OXPHOS genes or adjusting the metabolites production in mitochondria, tumor growth can be restrained. For example, enhanced complex I activity increases NAD⁺/NADH to prevent metastasis and progression of cancers. In this review, we discussed mitochondrial function in cancer cell metabolism and specially explored the unique role of mitochondria in cancer stem cells and the tumor microenvironment. Targeting the OXPHOS pathway and mitochondria-related metabolism emerging as a potential therapeutic strategy for various cancers.

The Mitochondrial Unfolded Protein Response: Role in Cellular Homeostasis and Disease.

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Mitochondria are the sole semiautonomous organelles of the human cell and play a very important role in not only energy production but also in apoptosis, metabolism and cell signaling. They are also known to be major producers of ROS and RNS free radicals during ATP production. These free radicals in excessive amount may damage the mitochondrial DNA as well as proteome resulting in accumulation of misfolded proteins which may prove deleterious to their functioning and are known to be involved in disease pathology. To maintain healthy proteome, mitochondria have developed as quality control machinery in semiautonomous manner, where cellular proteins such as proteases and heat shock proteins are used for quality control. The present review discusses various aspects of mitochondrial protein quality control operating at outer or inner membrane as well as intermembranal space. The various pathways involved in mitochondrial unfolded protein response have been discussed along with their implications in cancer and various neurodegenerative diseases.

Effects of exogenous bisphenol A on the function of mitochondria in root cells of soybean (*Glycine max* L.) seedlings.

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Bisphenol A (BPA), a contaminant of emerging concern, can affect plant root growth by changing various physiological processes. Mitochondria are critical organelles that produce energy for growth. However, how BPA affects the function and ultrastructure of mitochondria and then plant root growth remains unclear. Here, we evaluated the lethality of BPA to root tip cells, investigated the energy production process of mitochondria, observed mitochondrial ultrastructure, and measured reactive oxygen species (ROS) and lipid peroxidation levels in mitochondria of soybean seedlings roots exposed to exogenous BPA. We found that low-dose BPA (1.5 mg/L) exposure induced limited toxicity in root tip cells, increased the activities of key enzymes (citrate synthase, succinate dehydrogenase, malate dehydrogenase and cytochrome C oxidase) involved in tricarboxylic acid cycle and oxidative phosphorylation, promoted adenosine triphosphate (ATP) synthesis, and increased ROS production in mitochondria. Higher doses of BPA (6.0, 17.2 mg/L) exposure caused massive cell death in root tips, decreased the above key enzyme activities and ATP production, and destroyed mitochondrial ultrastructure; meanwhile, these doses also significantly increased mitochondrial ROS and membrane lipid peroxidation levels. In conclusion, we found that mitochondria were significant subcellular sites through which BPA can damage plant roots. BPA-induced excessive ROS destroyed mitochondrial ultrastructure and inhibited key enzyme activities in energy production, resulting in decreased ATP synthesis and cell death in root tips. Our results demonstrated the effects of BPA on mitochondrial function and structure in plant root cells, providing new insights into understanding the underlying mechanisms of BPA affecting plant root growth.

Mitochondria: participation to infertility as source of energy and cause of senescence.

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Mitochondria is a powerhouse organelle involved in ATP synthesis, calcium signaling, reactive oxygen species (ROS) by oxidative stress production, cell cycle arrest via apoptosis and sex steroid hormones biosynthesis. Improvement of sperm parameters such as motility, capacitation, acrosome reaction, and oocyte interaction, involve regulation of ROS levels by the mitochondria. In human, the relation between the quantitative level of mitochondrial DNA (mtDNA), oocyte cytoplasm maturation and fertilization

potential, is not clear. It has been hypothesized that oocytes without sufficient wild type mtDNA and therefore able to generate ATP, would not normally be ovulated. This is reflected in the low numbers of mtDNA observed in degenerate oocytes obtained through super ovulation protocols during assisted reproductive technology programs. Different theories place mitochondria in a central role of oxidative damage to cells and tissues related to infertility declining and aging. Mitochondria-dependent apoptosis seems to be responsible for the pre and post-natal decline in germ cells, embryo development, implantation failure, and miscarriages.

Complex formation and turnover of mitochondrial transporters and ion channels.

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Mitochondria are responsible for many vital cellular functions in eukaryotic cells, such as ATP production, steroid synthesis and prosthetic group biogenesis. The vital functions of mitochondria are possible due to the compartmental nature of this organelle. Mitochondria form a dynamic network that can exist as a network throughout a cell or as distinct individual structures. Mitochondria are also composed of two membranes, an inner and outer membrane. The inner mitochondrial membrane (IMM) is significantly larger than the outer membrane and must fold upon itself to be contained within the outer mitochondrial membrane (OMM). These folds are known as cristae. Altogether these different membrane compartments specialize in different functions of the mitochondria. The OMM is responsible for passage of small metabolites into and out of the mitochondria while excluding macromolecules. The IMM is a highly selective barrier between the solutes of the cytosol and those within the mitochondrial matrix. Cristae specialize in oxidative phosphorylation. The functions of these membranes are afforded by membrane proteins that are able to transport specific solutes. The appropriate localization, assembly into multi-subunit protein complexes, and wild-type function of these membrane proteins therefore is vital for mitochondria to maintain appropriate function and support cellular survival. This review will address the composition and functions of mitochondrial membrane localized multi-subunit protein complexes along with how these proteins undergo degradation to maintain homeostatic functions of mitochondria in the context of mitochondria specific transporters and ion channels. Due to the large number of known mitochondrial membrane transporters and ion channels this review will focus on the topics presented at the Mitochondrial Ion Channels and Transporters Symposium hosted by the New York University College of Dentistry in September 2015 in honor of Casey Kinnally.

Advances in myocardial energy metabolism: metabolic remodelling in heart failure and beyond.

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The very high energy demand of the heart is primarily met by adenosine triphosphate (ATP) production from mitochondrial oxidative phosphorylation, with glycolysis providing a smaller amount of ATP production. This ATP production is markedly altered in heart failure, primarily due to a decrease in mitochondrial oxidative metabolism. Although an increase in glycolytic ATP production partly compensates for the decrease in mitochondrial ATP production, the failing heart faces an energy deficit that contributes to the severity of contractile dysfunction. The relative contribution of the different fuels for mitochondrial ATP production dramatically changes in the failing heart, which depends to a large

extent on the type of heart failure. A common metabolic defect in all forms of heart failure [including heart failure with reduced ejection fraction (HFrEF), heart failure with preserved EF (HFpEF), and diabetic cardiomyopathies] is a decrease in mitochondrial oxidation of pyruvate originating from glucose (i.e. glucose oxidation). This decrease in glucose oxidation occurs regardless of whether glycolysis is increased, resulting in an uncoupling of glycolysis from glucose oxidation that can decrease cardiac efficiency. The mitochondrial oxidation of fatty acids by the heart increases or decreases, depending on the type of heart failure. For instance, in HFpEF and diabetic cardiomyopathies myocardial fatty acid oxidation increases, while in HFrEF myocardial fatty acid oxidation either decreases or remains unchanged. The oxidation of ketones (which provides the failing heart with an important energy source) also differs depending on the type of heart failure, being increased in HFrEF, and decreased in HFpEF and diabetic cardiomyopathies. The alterations in mitochondrial oxidative metabolism and glycolysis in the failing heart are due to transcriptional changes in key enzymes involved in the metabolic pathways, as well as alterations in redox state, metabolic signalling and post-translational epigenetic changes in energy metabolic enzymes. Of importance, targeting the mitochondrial energy metabolic pathways has emerged as a novel therapeutic approach to improving cardiac function and cardiac efficiency in the failing heart.

The *Toxoplasma gondii* homolog of ATPase inhibitory factor 1 is critical for mitochondrial cristae maintenance and stress response.

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The production of energy in the form of ATP by the mitochondrial ATP synthase must be tightly controlled. One well-conserved form of regulation is mediated via ATPase inhibitory factor 1 (IF1), which governs ATP synthase activity and gene expression patterns through a cytoprotective process known as mitohormesis. In apicomplexans, the processes regulating ATP synthase activity are not fully elucidated. Using the model apicomplexan *Toxoplasma gondii*, we found that knockout and overexpression of TgIF1, the structural homolog of IF1, significantly affected gene expression. Additionally, TgIF1 overexpression resulted in the formation of a stable TgIF1 oligomer and increased the presence of higher order ATP synthase oligomers. We also show that parasites lacking TgIF1 exhibit reduced mitochondrial cristae density, and that while TgIF1 levels do not affect growth in conventional culture conditions, they are crucial for parasite survival under hypoxia. Interestingly, TgIF1 overexpression enhances recovery from oxidative stress, suggesting a mitohormetic function. In summary, while TgIF1 does not appear to play a role in ATP synthase regulation under conventional growth conditions, our work uncovers its potential role in adapting to the stressors faced by *T. gondii* and other apicomplexans throughout their intricate life cycles.

Mitochondrial protein synthesis: efficiency and accuracy.

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The mitochondrial genetic system is responsible for the production of a few core-subunits of the respiratory chain and ATP synthase, the membrane protein complexes driving oxidative phosphorylation (OXPHOS). Efficiency and accuracy of mitochondrial protein synthesis determines how efficiently new OXPHOS complexes can be made. The system responsible for expression of the mitochondrial-encoded

subunits developed from that of the bacterial ancestor of mitochondria. Importantly, many aspects of genome organization, transcription, and translation have diverged during evolution. Recent research has provided new insights into the architecture, regulation, and organelle-specific features of mitochondrial translation. Mitochondrial ribosomes contain a number of proteins absent from prokaryotic ribosomes, implying that in mitochondria, ribosomes were tailored to fit the requirements of the organelle. In addition, mitochondrial gene expression is regulated post-transcriptionally by a number of mRNA-specific translational activators. At least in yeast, these factors can regulate translation in respect to OXPHOS complex assembly to adjust the level of newly synthesized proteins to amounts that can be successfully assembled into respiratory chain complexes. Mitochondrial gene expression is determining aging in eukaryotes, and a number of recent reports indicate that efficiency of translation directly influences this process. Here we will summarize recent advances in our understanding of mitochondrial protein synthesis by comparing the knowledge acquired in the systems most commonly used to study mitochondrial biogenesis. However, many steps have not been understood mechanistically. Innovative biochemical and genetic approaches have to be elaborated to shed light on these important processes.

Sigma-1 Receptor Promotes Mitochondrial Bioenergetics by Orchestrating ER Ca²⁺ Leak during Early ER Stress.

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The endoplasmic reticulum (ER) is a complex, multifunctional organelle of eukaryotic cells and responsible for the trafficking and processing of nearly 30% of all human proteins. Any disturbance to these processes can cause ER stress, which initiates an adaptive mechanism called unfolded protein response (UPR) to restore ER functions and homeostasis. Mitochondrial ATP production is necessary to meet the high energy demand of the UPR, while the molecular mechanisms of ER to mitochondria crosstalk under such stress conditions remain mainly enigmatic. Thus, better understanding the regulation of mitochondrial bioenergetics during ER stress is essential to combat many pathologies involving ER stress, the UPR, and mitochondria. This article investigates the role of Sigma-1 Receptor (S1R), an ER chaperone, has in enhancing mitochondrial bioenergetics during early ER stress using human neuroblastoma cell lines. Our results show that inducing ER stress with tunicamycin, a known ER stressor, greatly enhances mitochondrial bioenergetics in a time- and S1R-dependent manner. This is achieved by enhanced ER Ca²⁺ leak directed towards mitochondria by S1R during the early phase of ER stress. Our data point to the importance of S1R in promoting mitochondrial bioenergetics and maintaining balanced H₂O₂ metabolism during early ER stress.
